

and control the device.

To operate the device remotely, install the Arduino IoT Remote app

from the Google Play store and sign in using the same Arduino account.

The system can also be controlled via

the Arduino IoT cloud dash-board in a web browser. The setup process involves the steps mentioned below.

To create a thing and variables in Arduino IoT Cloud:

- Log in to

Arduino IoT Cloud and create a new thing.

- Connect the IndusBoard Coin to a computer using a USB cable and set it as the associated device.

- Select ESP32 as the device type and ESP32S2 dev module as the board.

- Save the generated device key, as it will be required later for authentication.

- Create four cloud variables named forward, backwards, left, and right, each with the Boolean (bool) data type.

- Set the permissions for all vari-

ables to Read & Write and the update policy to On change. Fig. 2 shows the Arduino IoT Cloud configuration used for this system.

To configure the device, follow the steps mentioned below:

- Select the target board (ESP32S2 dev module).

- Enter the Wi-Fi SSID and password under the network settings section.

- Save the configuration.

Next, generate and modify the sketch through following steps:

- Open the sketch tab in Arduino IoT Cloud to automatically generate the system code.

TABLE 1
BILL OF MATERIALS

Material	Description	Quantity
N20 DC gear motor	Motor 1, motor 2, motor 3, motor 4	4
N20 wheels	Wheels for motors	4
IndusBoard Coin	Controller board (programmed with code)	1
Connector wires	Interconnection wiring	As required
Switch	Power on/off control	1
L293D	Motor driver IC	1
6V battery	Power supply	1

sketch_may28a.ino

```

1  #include "thingProperties.h"
2  #include "arduino_secrets.h"
3  int indicator_led=33;
4  int first_switch=38;
5  int second_switch=39;
6  int third_switch=40;
7  int fourth_switch=41;
8  void setup() {
9    // Initialize serial and wait for port to open:
10   Serial.begin(9600);
11   // This delay gives the chance to wait for a Serial Monitor without block
12   delay(500);
13   pinMode(first_switch, OUTPUT);
14   pinMode(second_switch, OUTPUT);
15   pinMode(third_switch, OUTPUT);
16   pinMode(fourth_switch, OUTPUT);
17   pinMode(indicator_led, OUTPUT);
18   // Defined in thingProperties.h
19   initProperties();
20   // Connect to Arduino IoT Cloud
21   ArduinoCloud.begin(ArduinoIoTPreferredConnection);
22   /*
23
```

Fig. 3: Code snippet

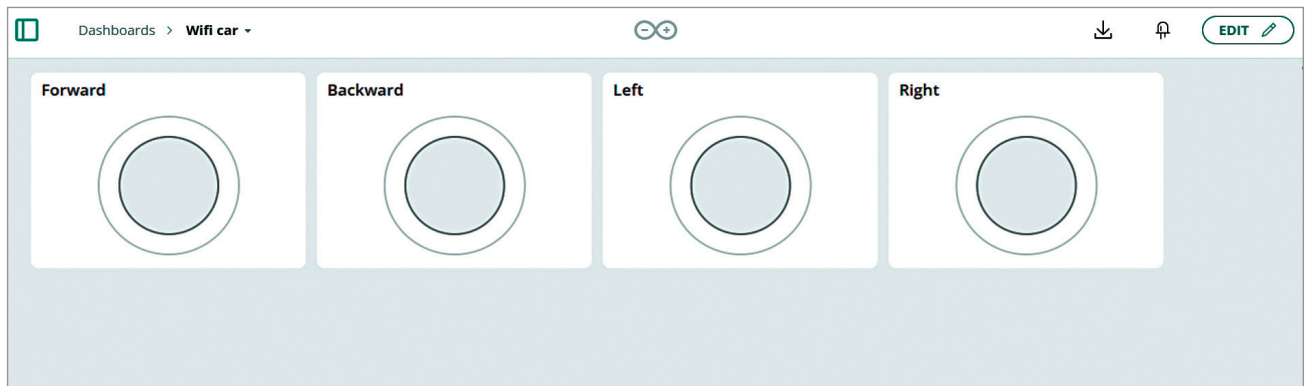


Fig. 4: IoT cloud dashboard